Semester V **B. Tech CSE (AI)**

BCSAI 505: FORMAL LANGUAGE AND AUTOMATA

3L + 0T + 0P + 3C **MM 100**

Unit 1: Finite Automata & Regular Expression

Basic Concepts of finite state system, Deterministic and non-deterministic finite automation and designing regular expressions, relationship between regular expression & Finite automata minimization of finite automation mealy & Moore Machines.

Unit 2: Regular Sets Of Regular Grammars

Basic Definition of Formal Language and Grammars, Regular Sets and Regular Grammars, closure proportion of regular sets, Pumping lemma for regular sets, decision Algorithms for regular sets, Myhell_Nerod Theory & Organization of Finite Automata.

Unit 3: Context Free Languages& Pushdown Automata

Context Free Grammars – Derivations and Languages – Relationship between derivation and derivation trees – ambiguity – simplification of CEG – Greiback Normal form – Chomsky normal forms – Problems related to CNF and GNF Pushdown Automata: Definitions – Moves – Instantaneous descriptions – Deterministic pushdown automata – Pushdown automata and CFL - pumping lemma for CFL - Applications of pumping Lemma.

Unit 4: Turing Machines

Turing machines – Computable Languages and functions – Turing Machine constructions – Storage in finite control – multiple tracks – checking of symbols – subroutines – two way infinite tape. Undecidability: Properties of recursive and Recursively enumerable languages – Universal Turing Machines as an undecidable problem – Universal Languages – Rice's Theorems.

Unit 5: Linear Bounded Automata Context Sensitive Language

Chomsky Hierarchy of Languages and automata, Basic Definition& descriptions of Theory & Organization of Linear bounded Automata Properties of context-sensitive languages.

Text/Reference Books:

1. Hopcroft, Ullman, "Introduction to Automata Theory, Language and Computation",Nerosa Publishing House
2. K.L.P. Mishra and N.Chandrasekaran, "Theory of Computer Science (Automata,Languages and Computation)", PHI
3. Martin J. C., "Introduction to Languages and Theory of Computations", TMH
4. Hopcroft, Ullman, "Introduction to Automata Theory, Languages and Computation", Pearson Education
5. Papadimitrou, C. and Lewis, C.L., "Elements of the Theory of Computation", PHI